

27.	C	Diffraction is a phenomenon exhibited by waves. Electron undergoing diffraction shows that it has wave property.
28.	B	Energy (= eV) transferred to the electron and proton is the same, since they have the same magnitude of charge, so since $E = \frac{p^2}{2m}$, their momentum is given by $p = \sqrt{2mE}$, so their associate wavelength is $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}}$. Hence $\lambda \propto \frac{1}{\sqrt{m}} \text{ or } \frac{\lambda_p}{\lambda_e} = \sqrt{\frac{m_e}{m_p}} = \sqrt{\frac{9.11 \times 10^{-31}}{1.67 \times 10^{-27}}} = 0.0234$
29.	C	Original count rate per minute due to source = $522 - 24 = 508$ per minute